Energy Reports 12 (2024) 1449–1463

Contents lists available at ScienceDirect

Energy Reports

journal homepage: www.elsevier.com/locate/egyr

Research paper

Fault detection and synchronization control in hybrid DC/AC microgrids using grid-forming inverter DC-link controller

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ARTICLE INFO

Keywords:
DC-link fault detection
Virtual synchronous generator (VSG) control
Grid-forming (GFM) inverters
Hybrid DC/AC microgrid
System stability

ABSTRACT

This paper introduces a DC-link fault detection and synchronization control strategy for grid-forming inverters in hybrid DC/AC microgrids, aiming to bolster system stability and reliability. The proposed control scheme seamlessly integrates DC-link fault detection with virtual synchronous generator (VSG) control, facilitating inertia emulation and synchronization without the need for a phase-locked loop (PLL). The DC bus controller is designed to swiftly identify faults triggered by power imbalances, such as load fluctuations or generator integration, and dynamically adjust the power reference for the VSG controller to ensure inertia emulation. The Ziegler-Nichols method is utilized for meticulous tuning of the DC-link controller gains, while the modified swing equation governs VSG inertia and synchronization. MATLAB/Simulink demonstrate the superior performance of the proposed method compared to traditional GFM control strategies. Specifically, the proposed controller improved the frequency nadir to 59.98 Hz compared to 59.75 Hz and showed an enhanced power step response of 7.9×10ˆ5 W compared to 6.9×10ˆ5 W for conventional control method. The proposed GFM control also eliminated the 10×10ˆ6 W switching spike observed with conventional controllers. These findings highlight the potential of this fault-controller approach to advance the reliability and efficiency of hybrid DC/AC microgrids.

1. Introduction

Optimizing power system stability is crucial for the successful integration of renewable energy sources into the electric grid (Alam et al., 2020; Hong et al., 2021). Grid-forming (GFM) inverters play a vital role in maintaining stable grid frequency and voltage, especially in low inertia power systems with a high penetration of inverter-based resources (Matevosyan et al., 2019; Kroposki et al., 2017; Zhou et al., 2023). However, conventional phase-locked loop (PLL) controllers, commonly used for synchronization in grid-connected inverters, can face challenges in weak AC systems or grids with a high share of inverters (Li et al., 2024; Zou et al., 2024; Zhang and Schuerhuber, 2023). As a result, significant research has been devoted to investigating alternative GFM control strategies such as droop control (Liao et al., 2019; Liu et al., 2016), virtual synchronous generators (VSGs) (Gao et al., 2023; Feng and Liu, 2022), and various DC-link synchronization methods (Zhao et al., 2022; Zhou et al., 2021). VSG control, which emulates the inertial behaviour of synchronous generators, has shown particular promise but often relies on AC frequency/voltage measurements and controllers (Gao et al., 2023; Li and Chen, 2018). Some work has explored using DC-link dynamics for VSG synchronization (Zhao et al., 2022; Zheng et al., 2021) but this has typically focused only on the DC-side dynamics without fully integrating AC and DC control aspects. In parallel, researchers have investigated techniques for fault detection, isolation, and control reconfiguration for GFM inverters (Zhang and Schuerhuber, 2023; Fan et al., 2022). Common methods include overcurrent protection via thresholds/virtual impedances (Fan et al., 2022; Qoria et al., 2020; Anon, 2020) and model-based diagnostics (Qoria et al., 2020). However, these approaches can be limited by the specificity of the fault signatures they detect and may not adequately address the dynamics of DC-link power imbalances. While prior work has made valuable advances, there remains a need for integrated control solutions that can robustly synchronize GFM inverters across a wide range of operating conditions while providing enhanced fault detection, isolation, and mitigation - especially for faults manifesting as DC-link power imbalances. This paper proposes a novel strategy that tightly integrates

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https://doi.org/10.1016/j.egyr.2024.07.025
Received 13 February 2024; Received in revised form 11 June 2024; Accepted 12 July 2024
Available online 25 July 2024

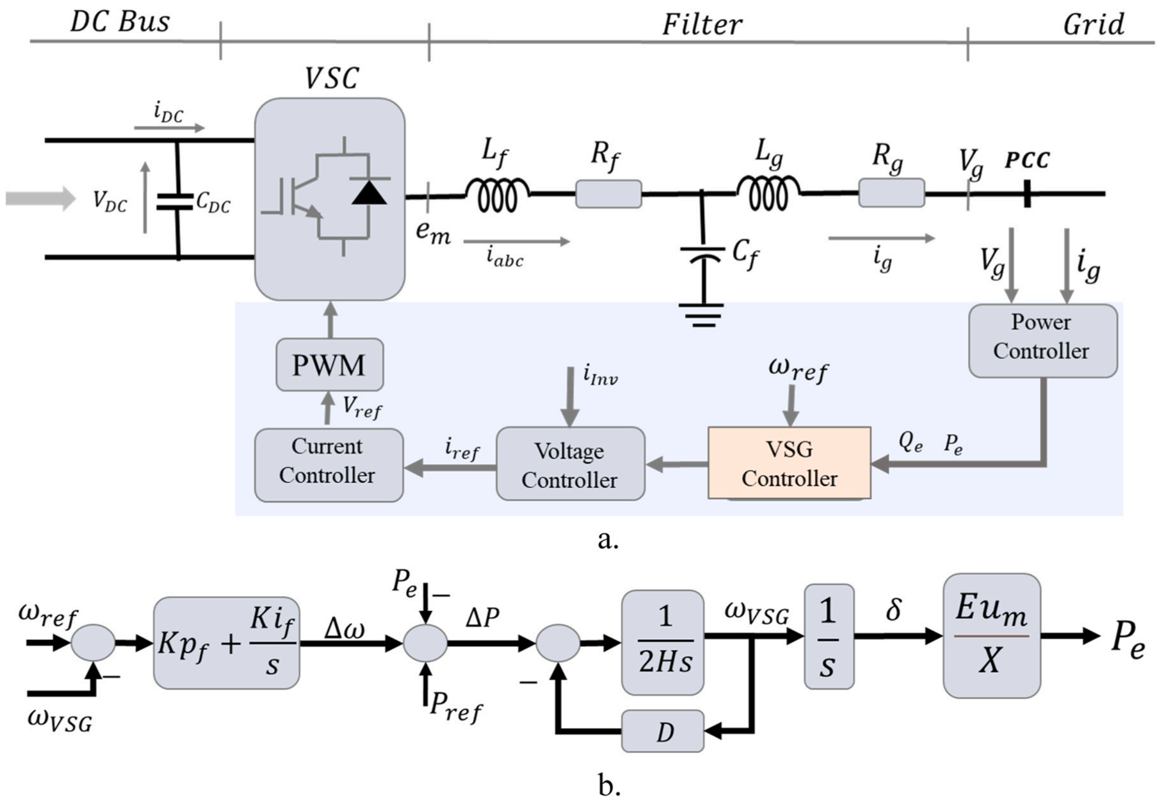


Fig. 1. General Configuration of convention GFM control. (a) Conventional GFM structure. (b) VSG synchronization control block.

Table 1
GFM inverter general parameters.

Parameter	Value	Parameter	Value
Rated power S_n	1 MVA	Grid resistance R_g	0.005 pu
Rated voltage V_n	32 kV	Grid inductance L_g	0.20 pu
Rated frequency f_n	60 Hz	DC bus Capacitance C_{dc}	3.45e−5 F
Inertia constant H	1 sec	DC bus Voltage v_{dc}	16 kV
Damping coefficient D	10	DC PI gains K_t, K_d	2, 16
Filter resistance R_f	0.005 pu	Inverter voltage and grid voltage u_m, E	0.98pu, 1pu
Filter inductance L_f	0.20 pu	AVR controller gains K_q, K_{p_Q}, K_{i_Q}	5.0, 50, 120
Filter Capacitance C_f	0.07 pu	Detection PI gain K_{p_d}, K_{i_d}	0.28, 14.7
Ref. Angular freq. $\omega_{ref} = \omega_0$	1pu	Governor PI gain K_{p_f}, K_{i_f}	15, 60
Grid SCR	20	Line Impedance in pu L_1, L_2, R_1, R_2	0.16, 0.15,0.03,0.026

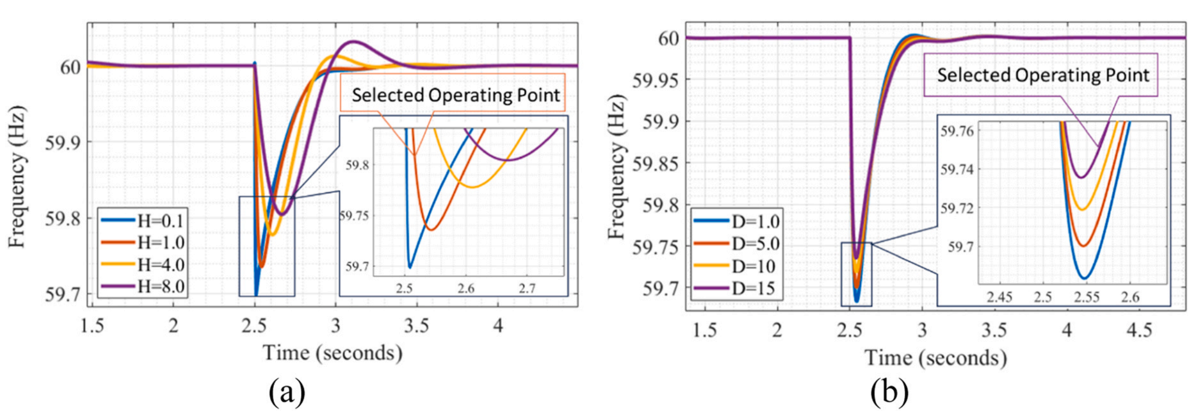


Fig. 2. Inertia and damping parameter settings. (a) Change in inertia constant H, with damping coefficient set at, D=15. (b) Change in damping coefficient D, with inertia coefficient set at H=1.0.

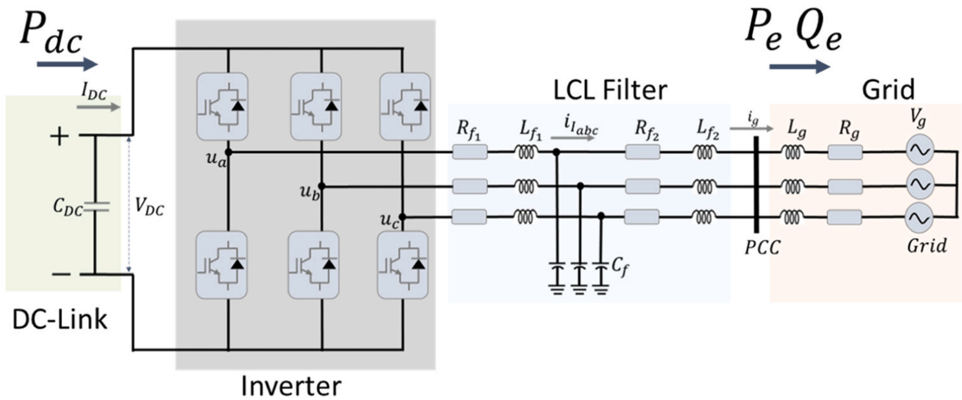


Fig. 3. Structural Configuration of power transfer through inverter from the DC side to the AC side.

DC-link fault detection and mitigation capabilities with VSG-based synchronization control leveraging both the DC and AC side dynamics of the system. By synthesizing techniques from DC-link control, VSG control, and power systems stability theory, the proposed approach is designed to provide fast fault detection without explicit thresholds, automatic calculation of appropriate power references for the VSG controller, and improved inertia response - thereby enhancing system stability and reliability for hybrid DC/AC microgrid applications. The key contributions of this paper are:

1. A DC-link controller that detects power imbalances and automatically generates a power reference for the VSG controller to provide inertia emulation.
2. Integration of the DC-link controller with the VSG synchronization mechanism to leverage both DC and AC side dynamics for enhanced stability.
3. Using the Ziegler-Nichols method to systematically tune the gains of the DC-link controller.

The rest of the paper is organized as follows: Section 2 provides a detailed examination of the essence of synchronization for connected generators, including conventional and proposed GFM synchronization methods. Section 3 analyzes the stability of these synchronization controllers. Section 4 focuses on simulation results and their discussion. Finally, the paper concludes with Section 5, summarizing key findings and suggesting directions for future research.

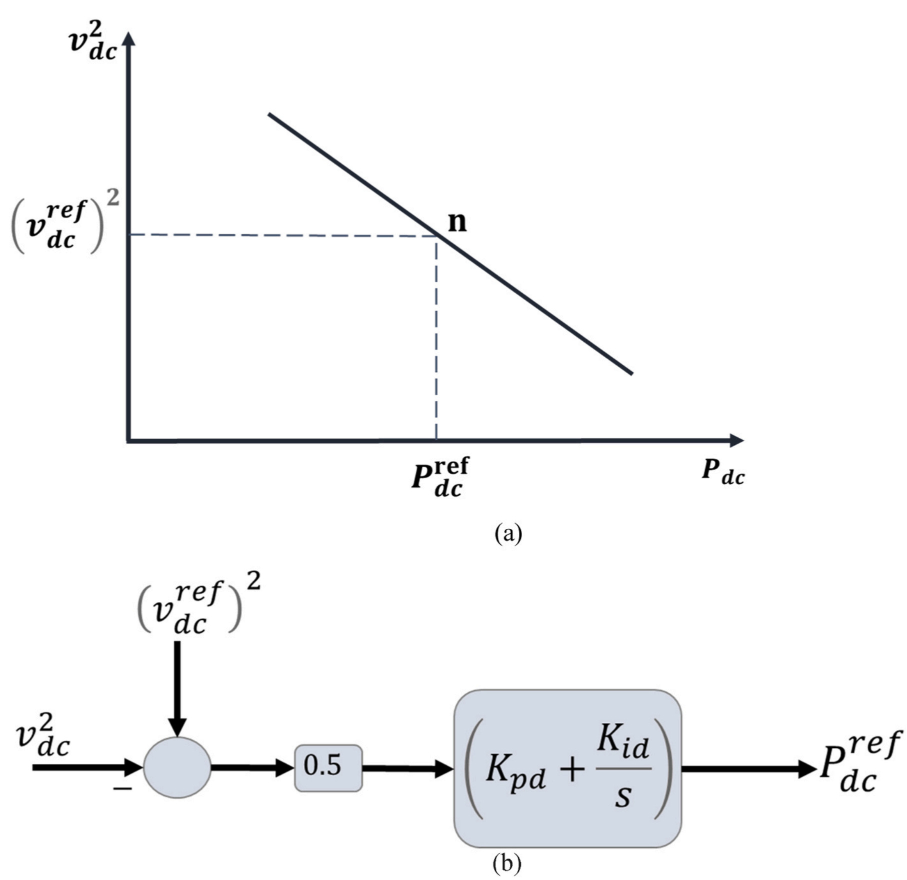


Fig. 4. The DC power reference control. (a) The DC drop control curve. (b)The DC power reference transfer block.

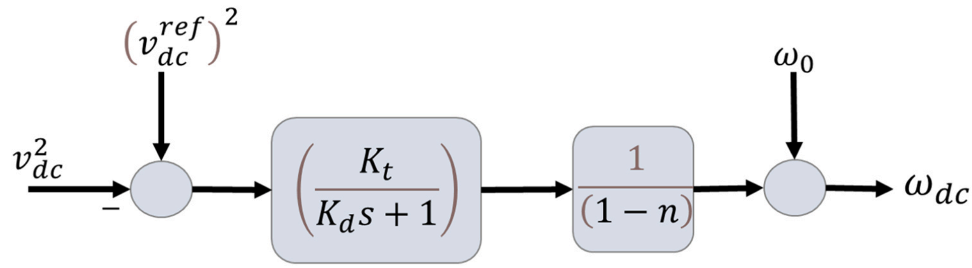


Fig. 5. DC angular speed reference controller.

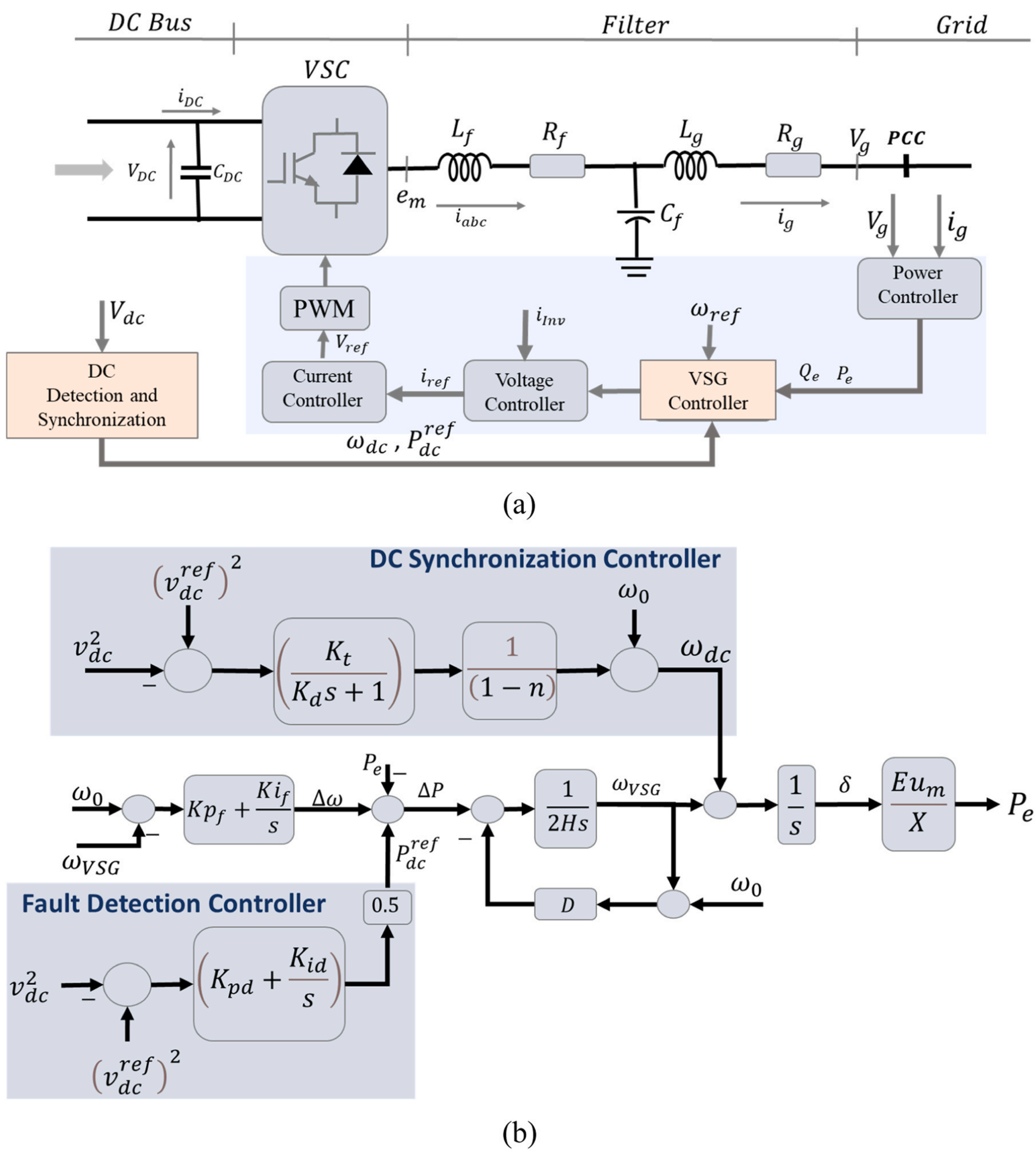


Fig. 6. The proposed GFM controller. (a) General configuration of the proposed GFM controller. (b) Proposed detection and synchronization controller scheme.

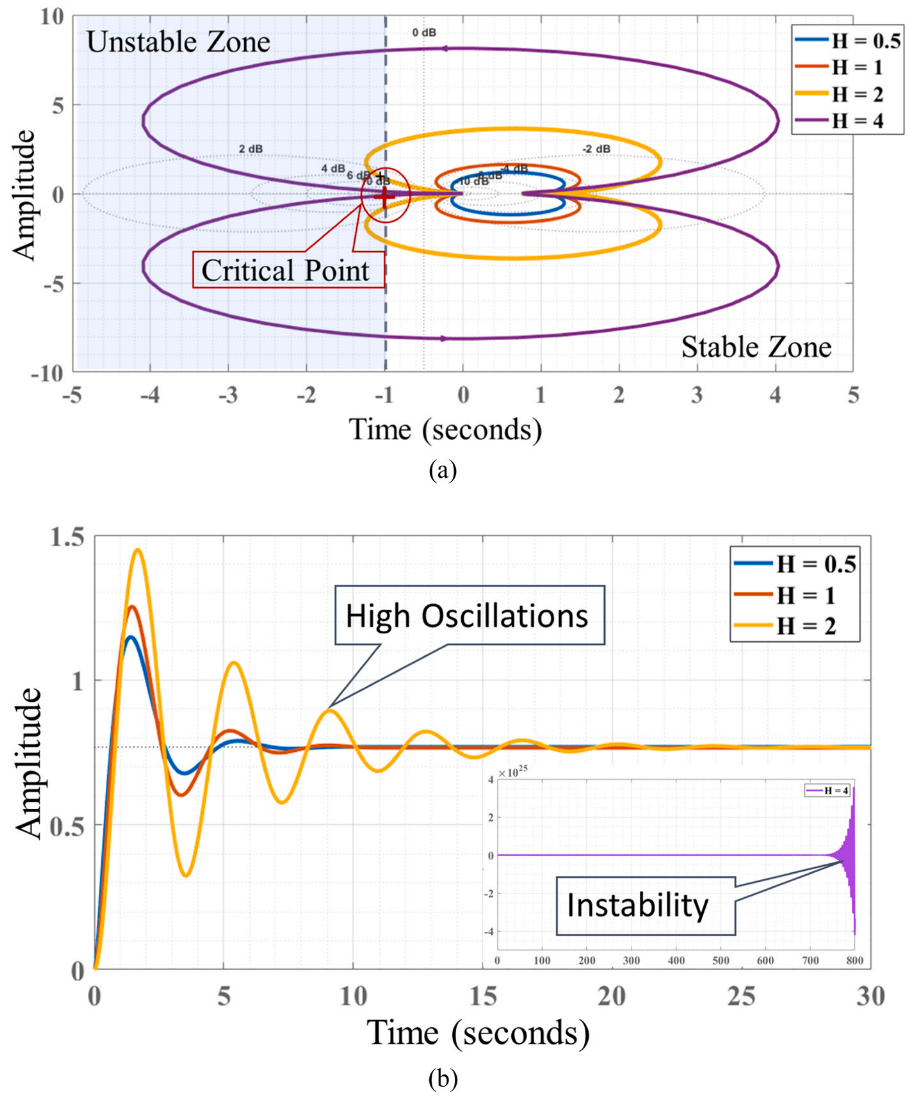


Fig. 7. Graphical analysis of conventional synchronization controller. (a). The Nyquist plot map (b) The Step Response plot.

2. The essence of synchronization

2.1. Conventional GFM synchronization control

GFM inverters synchronize with the grid by adjusting their active power output, similar to synchronous generators (SG). This allows them to stay synchronized even in low short circuit ratio (SCR) grids. However, in high SCR grids, GFM inverters may lose synchronism due to small phase differences causing large active power variations (Han et al., 2024; Liu et al., 2024).Therefore, robust damping control is needed for converters operating across varying SCR conditions. When operating parallel to a small SG, the GFM must emulate the SG's pole pairs to match their dynamics, as merely matching the moment of inertia (J) is insufficient. The equation expresses the relationship between the GFM and

$$\frac{J_{GFM}}{S_{b_{GFM}} p_{GFM}^2} = \frac{J_{SG}}{S_{b_{SG}} p_{SG}^2} \tag{1}$$

where $S_{b_{GFM}}$ and $S_{b_{SG}}$ are the base rated power of the GFM and SG, respectively. Here, p_{GFM} and p_{SG} represents the pole pairs. The number of rotor poles is a crucial parameter affecting the microgrid dynamics with both VSG and SG. From the SG swing equation (Mohammed et al., 2019; Ebinyu et al., 2023),

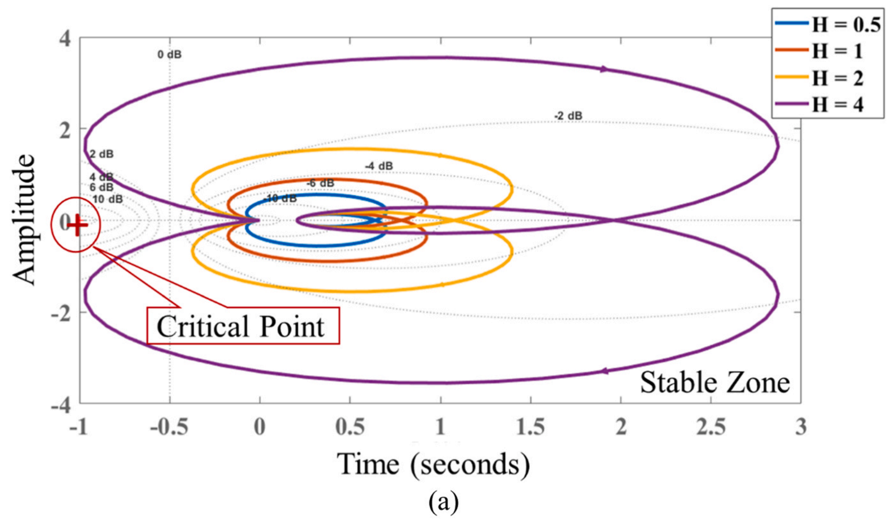
$$H\frac{d\omega_s}{dt} = P_{ref} - K_f\Delta\omega - P_e + D\Delta\omega \tag{2}$$

where P_{ref} is the power reference, P_e is the electromagnetic power, D is the damping coefficient, and K_f is the frequency constant given by $\frac{1}{R}$. R is the speed regulation factor whose value is 5 % (Feng and Liu, 2022; Shi et al., 2021; Zuo et al., 2021). A PI controller is used instead of the frequency constant ($K_f \rightarrow K_{p_f} + \frac{K_{i_f}}{s}$). The PI controller provides frequency restoration and a more robust speed regulatory control (Shi et al., 2021; Zhu et al., 2021). The Inertia constant $H$is often given as $H = \frac{1/2J\omega^2}{S_b}$. The swing equation with the PI speed regulator is given by

$$H\frac{d\omega_s}{dt} = P_{ref} - (K_{p_f} + \frac{K_{i_f}}{s})\Delta\omega - P_e + D\Delta\omega \tag{3}$$

The conventional GFM scheme and its synchronization control diagram are shown in Fig. 1.

From the isolated GFM configuration in Fig. 1 from where the inertia and the damping parameters are selected. The main inertia control function is given as, $\frac{1}{2Hs+D}$. The system parameters are shown on Table 1. To get the appropriate operating inertia, H and Damping D. The simulation result for this selection is shown in Fig. 2. The simulation is performed for a power step change at 2.5 seconds, the system frequency



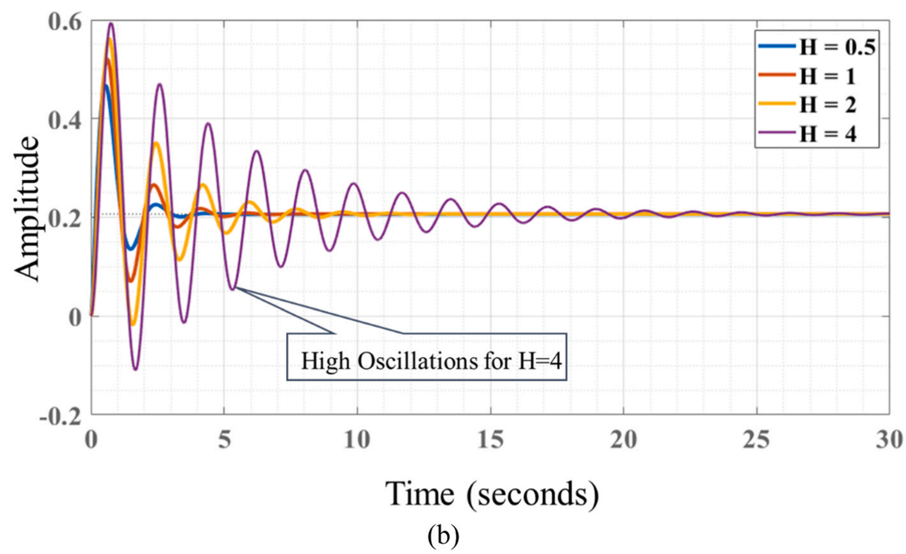


Fig. 8. Graphical analysis of the proposed synchronization controller. (a). The Nyquist plot map (b) The Step Response plot.

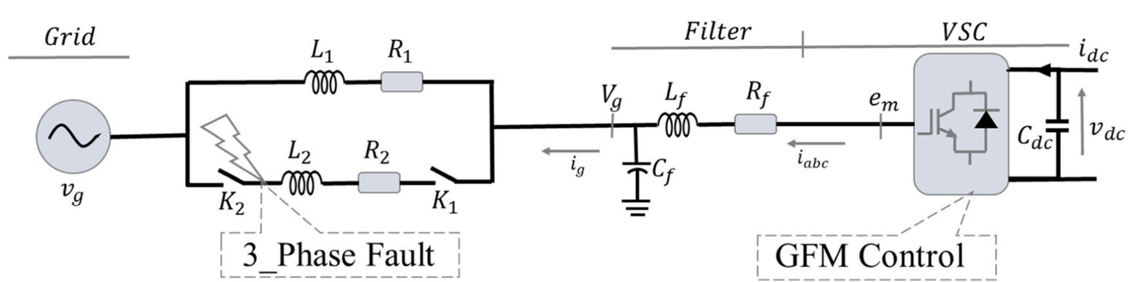


Fig. 9. Structural Configuration of a Single GFM Connected to the Grid.

response is analyzed for different inertia and damping values. From the analysis, the operating inertia and damping are selected as H=1.0 and D=15.

$$T_{(s)} = \frac{\Delta P_e}{\Delta \omega} = \left(\frac{K_{p_f} s + K_{i_f}}{2Hs^3 + \left(D + K_{p_f}\right) s^2 + K_{i_f} s} \right) \left(\frac{Eu_m}{X} \right) \tag{4}$$

In the VSG control block diagram Fig. 1, the transfer function $T_{(s)}$ for conventional GFM is based on the power-frequency droop principle, describing power changes due to frequency variations. The conventional GFM synchronization controller with PI speed regulation is a third-order controller and does not involve DC-link control or DC parameters for inertia power. This results in inadequate control over power input and inertia power. Thus, a new controller is proposed to address these issues.

2.2. The proposed GFM control model

Consider the Grid-connected inverter as shown in Fig. 3.

Neglecting switching and line power loss, the system power balance equation is given by ($P_s = P_{dc} + P_e$), where P_s is the source input power,

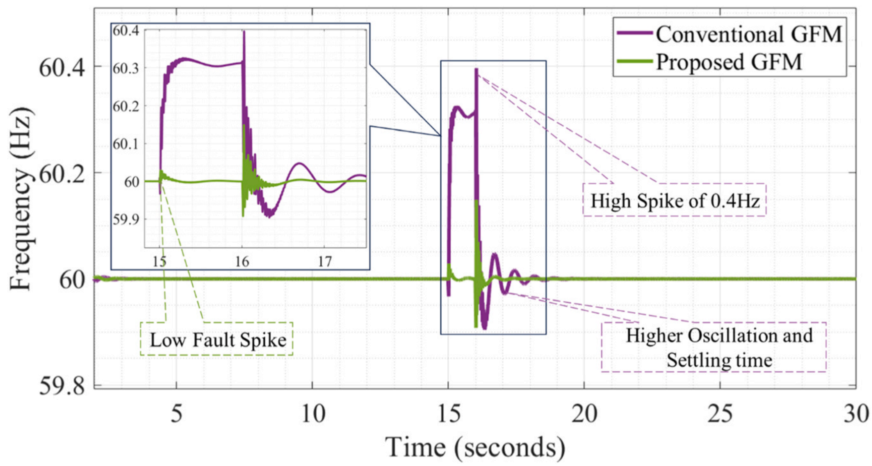


Fig. 10. Frequency Response to Three-Phase Line Fault.

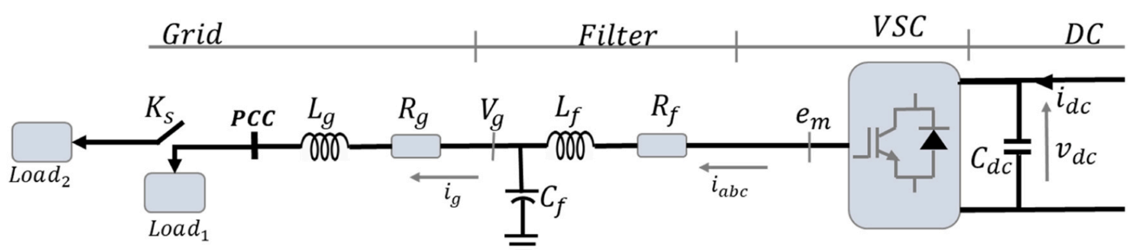


Fig. 11. Isolated GFM Configuration.

$P_{dc} = \frac{d}{dt}\left(\frac{1}{2}C_{dc}V_{dc}^2\right)$ the DC link power, and $P_e = (u_{abc}i_{I_{abc}})$ the output power. The state space dynamics equation of the system is given by.

$$P_s = \frac{d}{dt}\left(\frac{1}{2}C_{dc}V_{dc}^2\right) + (u_{abc}i_{I_{abc}}) \tag{5}$$

2.2.1. The proposed DC link fault detection and synchronization control

The oscillation of the AC active power resulting from unbalanced supply and demand conditions greatly affects the DC link voltage stability (Cupertino et al., 2019). The energy w_{dc} through the DC link, capacitor is given as $w_{dc} = \frac{C_{dc}V_{dc}^2}{2}$, where V_{dc} is the DC link voltage and C_{dc} the DC link capacitor. The DC link capacitor power balance is recalled from Eq. (6) as (Zarei et al., 2020; Vygoder et al., 2023)

$$\frac{d}{dt}\left(\frac{C_{dc}V_{dc}^2}{2}\right) = P_s - P_e \tag{6}$$

where P_s is the input source power and P_e the power at the inverter points of coupling. The DC voltage control is given as

$$i_{dc}^{ref} = (K_p^{dc} + \frac{K_i^{dc}}{s})(v_{dc}^{ref} - v_{dc}) \tag{7}$$

In which the DC PI controller proportional and integration gain are respectively K_p^{dc} and K_i^{dc}. To determine the optimal gains for the DC-link voltage controller (Eq. 8), the Ziegler-Nichols tuning method was applied. First, the DC microgrid was operated with the GFM inverter connected, and a step change in the load of 1 MW was applied. The DC-link voltage response was monitored, as shown in Fig. 19. The DC-link voltage exhibited an undamped oscillatory response with a peak-to-peak amplitude (A) of 0.15 kV and a period (T) of 0.2 seconds. Suppose the desired overshoot is less than 5 % using the Ziegler-Nichols tuning rules for this type of response (Muresan and De Keyser, 2022), the proportional gain (K_p^{dc}) was calculated as:$K_p^{dc} = 0.6 * C_{dc} * A = 0.6 * 3.45e - 5 * 0.15 = 3.11e - 6$. And the integral gain (K_i^{dc}) was determined as:$K_i^{dc} = 0.5 * C_{dc} * A/T = 0.5 * 3.45e - 5 * 0.15/0.2 = 2.59e - 5$. From the Ziegler-Nichols turning, the DC link parameters are calculated. The DC and voltage references respectively i_{dc}^{ref}, and v_{dc}^{ref}.The power imbalance between the source supply and the load demand is detected by the DC link voltage variation. The power transfer equation for an inverter can be expressed in terms of power at the DC side and power at the AC side, considering efficiency. The power equation relating to DC power P_{dc}, to AC power P_{ac} in a simple form is given by: $P_{ac} = nP_{dc}$. Here, n represents the efficiency of the conversion process from DC to AC power. The equation reflects how much of the DC power is effectively converted to AC power, considering losses in the conversion process. The DC power management is realized by the controller to obtain the power reference given by

$$v_{dc}^2 = \left(v_{dc}^{ref}\right)^2 - n \quad \left(P_{dc}^2 - \left(P_{dc}^{ref}\right)^2\right) \tag{8}$$

Where the DC power reference is controlled by

$$P_{dc}^{ref} = (K_{pd} + \frac{K_{id}}{s})\frac{\left(v_{dc}^{ref}\right)^2 - v_{dc}^2}{2} \tag{9}$$

The proposed DC link voltage controller scheme is shown in Fig. 4

The synchronization process ensures smooth and safe power transfer between distinct parts of the power grid. In a DC system, synchronization may involve aligning the voltage and phase of the DC source with the target DC system or with an AC system. The basic control strategy is.

$$\omega_{dc} = \omega_0 + \frac{K_t}{(1-n)K_d}\left[v_{dc}^2 - \left(v_{dc}^{ref}\right)^2\right] \tag{10}$$

The grid deviation under steady the controller provides voltage stability tracking with $n = 1 \rightarrow v_{dc} = v_{dc}^{ref}$

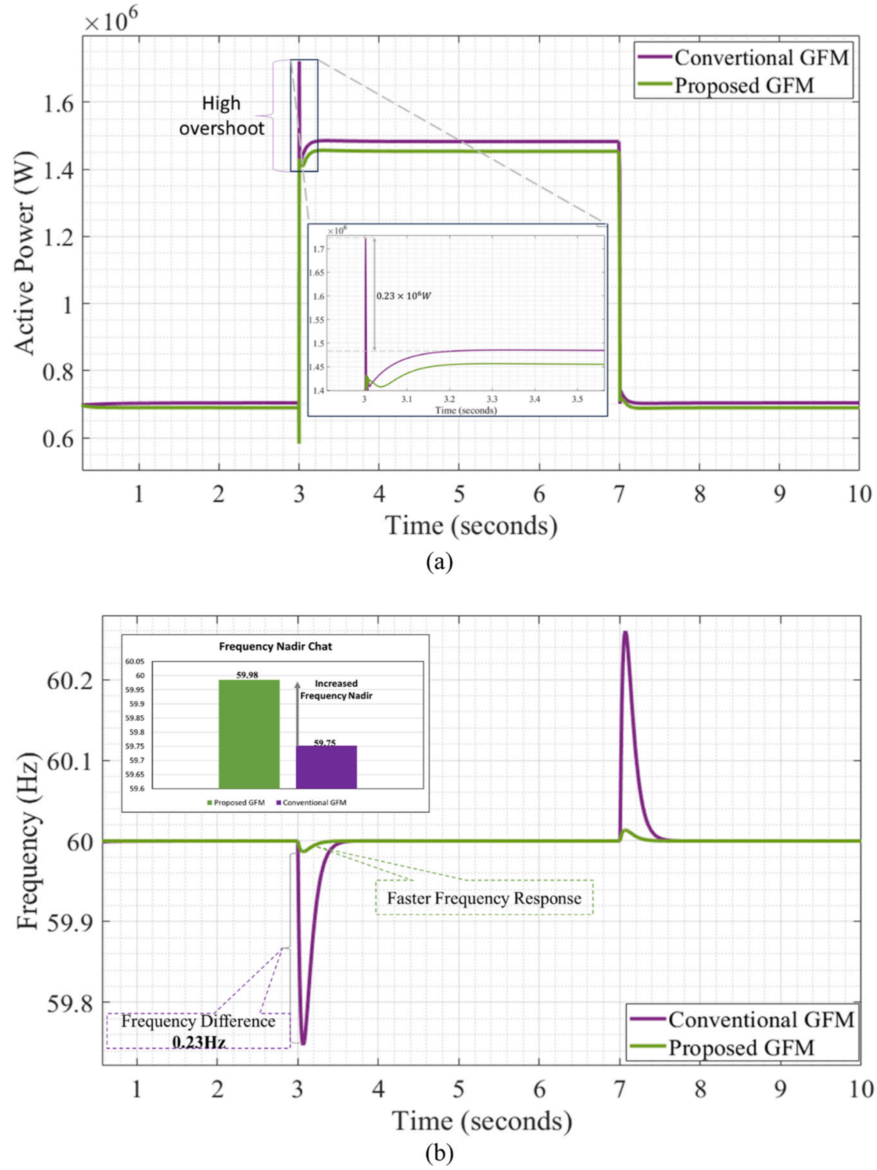


Fig. 12. System Response to Load Change. (a) Active Power Response. (b) Frequency Response.

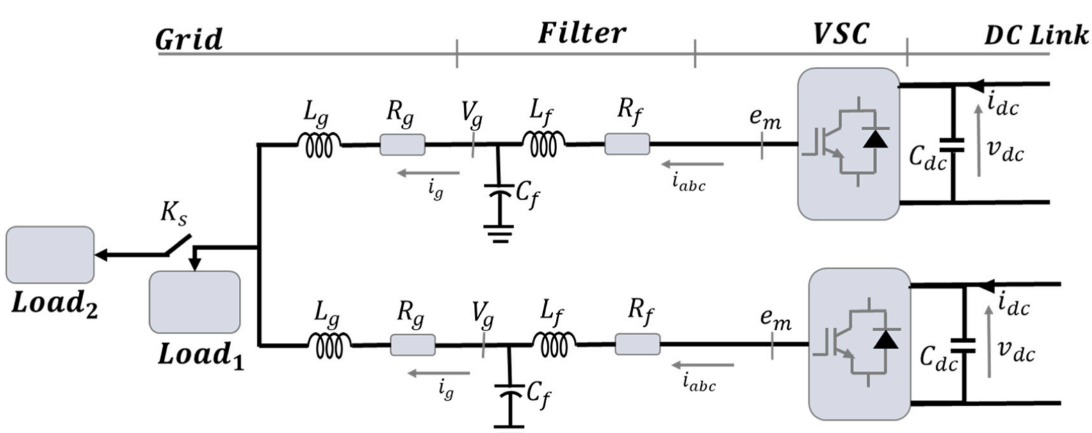


Fig. 13. Structure of the GFM parallel operation of conventional and the proposed controller.

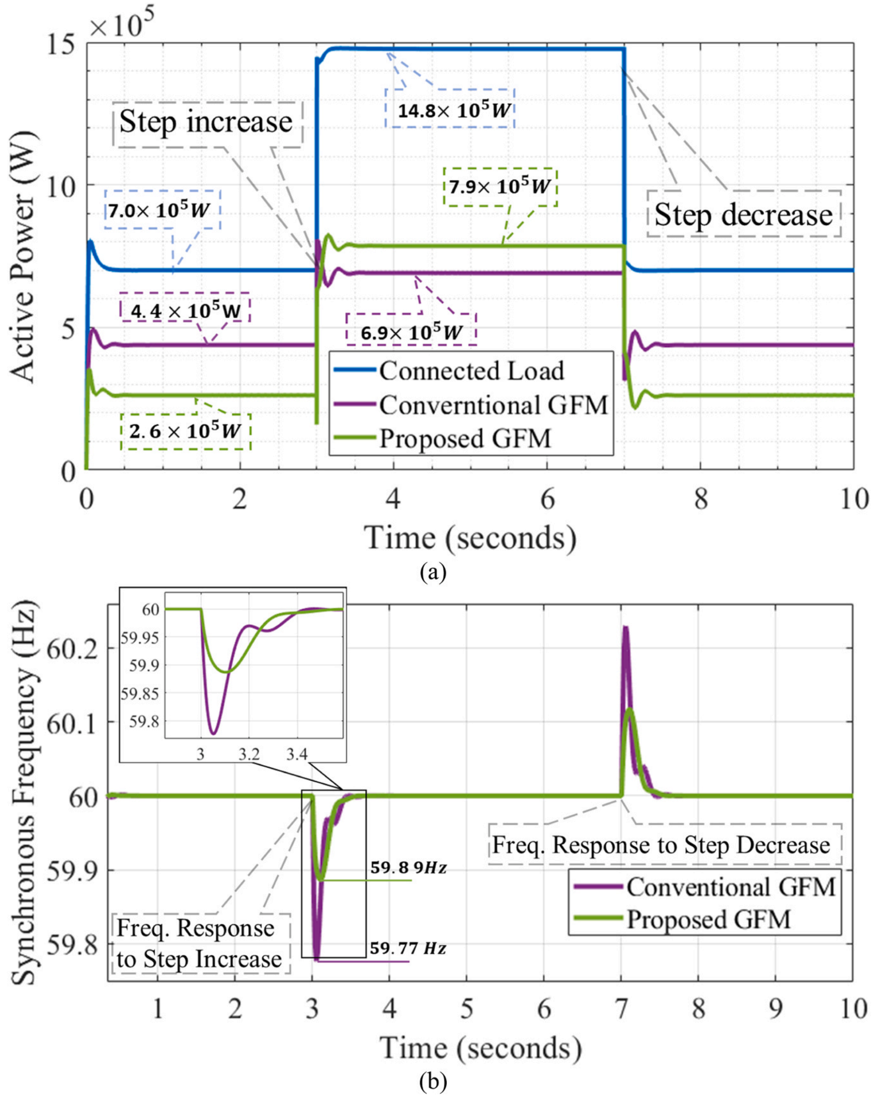


Fig. 14. Comparison of conventional and proposed GFM to step change. (a) Active power response. (b) Synchronous Frequency response.

2.2.2. *The proposed synchronization controller*

The proposed synchronization controller makes use of the DC link controller for fault detection, power, and phase referencing while the VSG controller is used for inertia control. The proposed scheme is shown in Fig. 6. From the control scheme, a more robust transfer function is obtained which describes the power transferred from the change in the DC link voltage to the injected active power change.

$$G_{(s)} = \frac{\Delta P_e}{\Delta\omega} = \left\{\left(\frac{K_{p_f}s + K_{i_f}}{2Hs^2 + (D + K_{p_f})s + K_{i_f}}\right) + \left(\frac{k_t}{(k_d s + 1)(1 - n)}\right)\right\}\left(\frac{Eu_m}{Xs}\right) \tag{11}$$

3. Controller stability analysis

3.1. *Conventional GFM controller stability analysis*

To analyse the stability of the given transfer function graphically, we need to examine the poles of the transfer function. We recalled the transfer function in Eq. (5) given by:

$$T(s) = \frac{\Delta Pe}{\Delta\omega} = \frac{K_{p_f}s + K_{i_f}}{2Hs^2 + (D + K_{p_f})s + K_{i_f}} \cdot \frac{Eu_m}{Xs} \tag{12}$$

Let us express the denominator in the standard form $D(s) = a_n s^n + a_{n-1}s^{n-1} + \ldots + a_1 s + a_0$. Then from $T(s)$ we have: $2Hs^3 + \left(D + K_{p_f}\right)s^2 + K_{i_f}s$

Now, we need to find the roots (poles) of this polynomial, which can be done by solving the equation $D(s)$=0. The poles determine the stability of the system.

To better simplify the analysis, the transfer function $T(s)$ is express it in the standard second-order form:

$$T(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \tag{13}$$

where $\omega_n$is the natural frequency and ζ is the damping ratio, we can match the coefficients: $2\zeta\omega_n = D + K_{p_f}$ and $\omega_n^2 = 2HK_{i_f}$from where we can obtain the natural frequency as: $\omega_n = \sqrt{\frac{K_{i_f}}{2H}}$ and the damping ratio as $\zeta = \frac{D + K_{p_f}}{2\sqrt{\frac{K_{i_f}}{2H}}}$. From the natural frequency, we notice that as the inertia increases, the natural frequency decreases by the square of its value and vice versa. On the other hand, the increase in inertia leads to a square

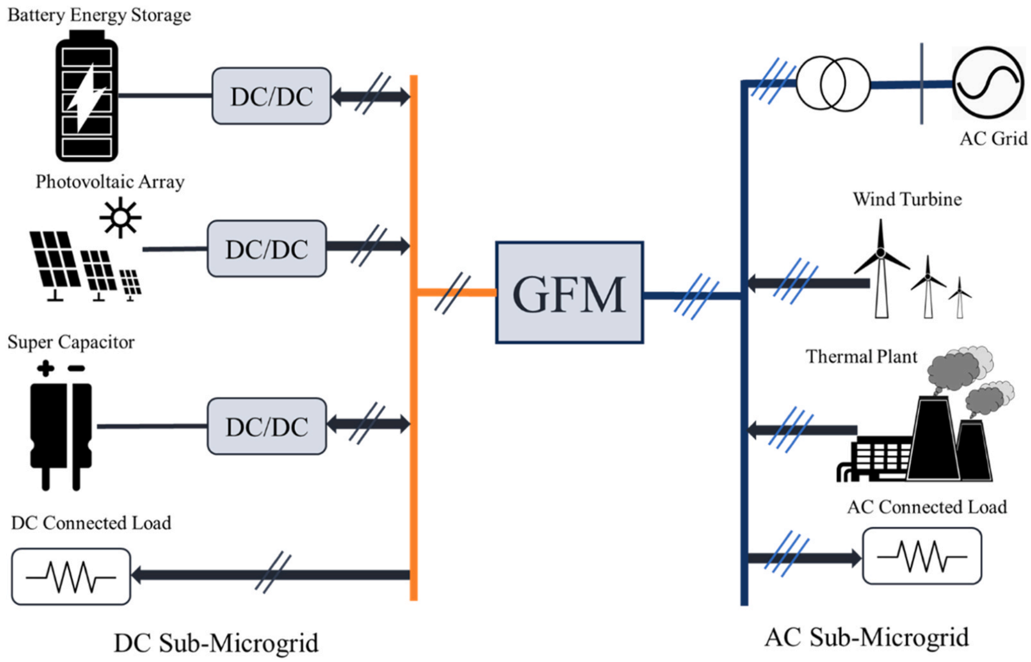


Fig. 15. Grid Structure. General Configuration of a DC/AC Hybrid microgrid interfaced through GFM inverters.

Table 2
DC/AC Microgrid Parameters.

DC Sub-Microgrid	
Generator	**Capacity**
PV System	2 MW
Battery	200 kWh
DC rated Voltage	800 V
AC Sub-Microgrid	
Generator	**Capacity**
AC Grid	2 MW
Wind Turbine	1.5 MW
Thermal Plant	0.5 MW
GFM inverter	1 MW
Rated Voltage	3 kV
Grid Frequency	60 Hz
Connected Loads	
DC load	0.5 kW
Initial AC Load	1 MW

increase in the damping ratio. This is shown graphically in Fig. 7 through the Nyquist plot and the step respond. The simulation and system parameters are shown in Table 1.

The Nyquist plot shown in Fig. 7 is divided into two zones (the stable and unstable zone) with the critical point at (-1,0) which show the boundaries between the stable and unstable zone. If the plots of the control system are located beyond the critical point toward the positive x-axis (stable zone), the system is stable and if the plots cross the critical point toward the negative x-axis (unstable zone), the system is unstable. Therefore, from the Nyquist plot, we observe that for smaller inertia values ($H = 0.5$ and $H = 1$) the plots lie within the stable zone and as the inertia value increases ($H = 2$ and $H = 4$) the plots are shifted to the unstable zone. This fact is further validated by the corresponding step response plot. From the step response, we observe that the increase in the inertia values increases the oscillations ($H = 2$) and at higher inertia the system becomes unstable ($H = 4$). From this analysis, we can conclude that, conventional GFM inverter control requires a low inertia value at steady state operations.

3.2. Proposed GFM controller stability analysis

The transfer function of the proposed controller is recalled from Eq. (12):

$$G(s) = \frac{\Delta Pe}{\Delta V_{dc}} = \left(\frac{(K_{p_f}s + K_{i_f})}{2HXs^2 + \left(D + K_{p_f}\right)s + K_{i_f}}\right)\cdot\frac{Eu_m}{Xs} + \frac{k_t}{(k_d s + 1)(1-n)}\cdot\frac{Eu_m}{Xs}$$

The poles of the characteristic equation of the transfer function is to be examined. The poles are the values of s that makes the denominator equal to zero. Let the denominator of the first part be denoted as $D_1(s)$ and the denominator of the second part as $D_2(s)$:

$$\begin{cases} D_1(s) = 2Hs^2 + \left(D + K_{p_f}\right)s + K_{i_f} \\ D_2(s) = (k_d s + 1)(1-n) \end{cases}$$

Now, these denominators are set to zero and solve for s. Solving these equations gives the poles of the transfer function. The stability of the system depends on the location of these poles in the complex plane:

- o If all poles have negative real parts, the system is stable.
- o If any pole has a positive real part, the system is unstable.

The denominator of the first term in standard form is: $D_1(s) = 2Hs^2 + \left(D + K_{p_f}\right)s + K_{i_f}$. From the standard form, $\omega_n^2 = 2H$ and $2\zeta\omega_n = (D + K_{p_f})$. So, the expressions for ω_n and ζ in terms of the given parameters are $\omega_n = \sqrt{2H}$ and $\zeta = \frac{D+K_{p_f}}{2\sqrt{2H}}$. Notice that the natural frequency is proportional to the square of the inertia and that the damping ratio value increases as the square of the inertia value increase. This analysis is graphically visualised through a Nyquist and step response plot as shown in Fig. 9.

The Nyquist plot in Fig. 8 illustrates how control stability changes with inertia. Compared to the conventional GFM control shown in Fig. 7, our proposed controller stays within the stable zone for all tested inertia values (H=0.5, 1, 2, and 4), avoiding the critical point (-1, 0). This is supported by step response tests, which reveal fewer oscillations at H=2 compared to the conventional method. These reduced oscillations improve power transfer quality and safety. In summary, our controller

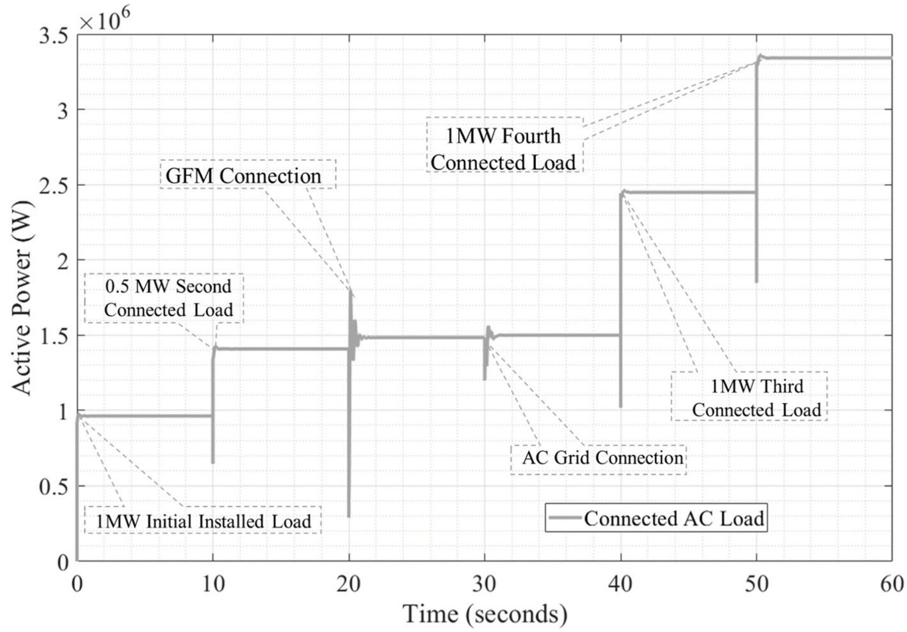


Fig. 16. Dynamics of AC-Connected Loads.

demonstrates superior stability compared to conventional controllers.

4. Simulation and results

4.1. Case 1: single GFM connected to the Grid.(Three Phase Fault)

In Case 1, the GFM system connects to the grid via two lines (Fig. 9). When a three-phase fault occurs on one line, breakers k_1 and k_2 open, isolating the line. The system's frequency response is compared between conventional and proposed GFM in Fig. 10. The fault begins at 15 seconds, lasting 0.5 seconds, with breakers opening and reclosing at 15 and 16 seconds, respectively. The comparison shows peak frequencies of 60.4 Hz (conventional) and 60.07 Hz (proposed GFM). This highlights the proposed GFM's superior stability, attributed to faster synchronization after faults compared to conventional methods.

4.2. Case 2: isolated GFM operation (Load step Change)

During isolated operations, load changes are common in power systems. Fig. 11 shows the configuration of the isolated GFM inverter. In Fig. 12, we compare frequency responses between the proposed GFM and conventional GFM. The simulation starts with an initial load of $0.7 \times 10^6 W$, maintaining system balance. At 3 seconds, a step increase introduces an additional load of $0.8 \times 10^6 W$ (Fig. 12(a)), causing a supply-demand imbalance perceived as a fault, resolved by 7 seconds. Comparing system frequency responses, the proposed GFM reaches a nadir frequency of 59.98 Hz, while the conventional GFM hits 59.75 Hz. This highlights the proposed GFM's faster response and enhanced stability.

4.3. Case 3: parallel operation of GFM

The operational performance of both the conventional and proposed GFM inverters will be compared during parallel operation with a load step change, as shown in Fig. 13. Fig. 14 displays the active power and synchronous frequency responses of both inverters, following identical specifications outlined in Table 1. In Fig. 14(a), the simulation starts with an initial load of $7.0 \times 10^5 W$, with the initial power allocation for the conventional and proposed GFM at $4.4 \times 10^5 W$ and $2.6 \times 10^5 W$, respectively. After 3 seconds, a step increase raises the load to 14.8×10^5 W. Consequently, the power allocation for the conventional GFM and proposed GFM shifts to $6.9 \times 10^5 W$ and $7.9 \times 10^5 W$, respectively. Initially, the proposed GFM supplies less power than its conventional counterpart. However, during the load step change, the proposed GFM supplies more power. This difference is attributed to the proposed DC link detection and synchronization controller, which operates based on DC power control. Upon detecting the step increase as a fault, the detection controller determines the required power to rectify the fault, resulting in power sharing dominance. This assessment is supported by the frequency response in Fig. 14(b), where the proposed controller yields a synchronous frequency nadir of 59.89 Hz, compared to 59.77 Hz for conventional GFM control, indicating improved performance.

4.4. Case 4: hybrid microgrid connection: DC/AC sub-microgrid interfaced by GFM inverter

The hybrid microgrid configuration, depicted in Fig. 15, comprises a DC sub-microgrid configuration consisting of PV systems and energy storage systems, including battery storage and supercapacitor storage. Supercapacitor storage is chosen for its rapid response compared to battery storage, a crucial feature for the proposed DC link fault detection and synchronization controller. Additionally, a DC load is integrated into the DC sub-microgrid. On the opposite side of the GFM inverter interface lies the AC sub-microgrid, comprising an AC-connected grid, a wind turbine, a thermal plant, and an AC-connected load. The parameters defining the DC/AC hybrid microgrid are detailed in Table 2. This paper primarily focuses on the microgrid configuration and its stability concerning generator connection. The dynamics of the DC side have not been discussed in this work. It is assumed that the DC/DC converters on the DC side maintain stable DC side voltage. The primary objective of this section is to maintain the stability enhancement facilitated by the proposed fault detection and synchronization control.

Utilizing MATLAB Simulink, a simulation is conducted based on the DC/AC microgrid configuration outlined in Fig. 15. To evaluate the performance of the proposed controller, initial testing involves connecting the conventional GFM inverters, followed by testing the proposed GFM. The simulation commences with the GFM disconnected, indicating that the DC sub-microgrid operates in islanded mode. The paper's focus lies on assessing the AC side's performance and its impact

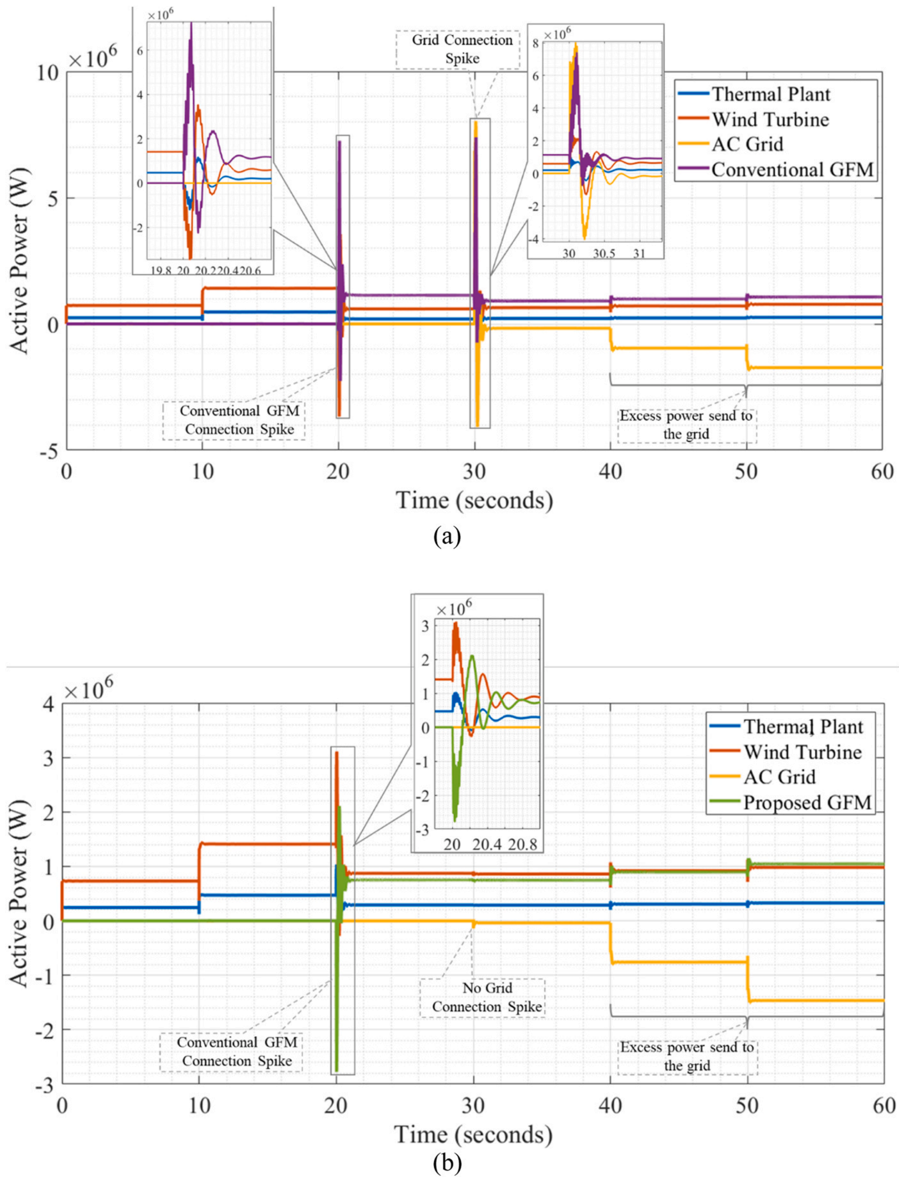


Fig. 17. Active Power Sharing in AC Sub-Grid: (a) Conventional GFM Connected System, (b) Proposed GFM Connected System.

on the DC side. On the AC side, the simulation begins with a 1 MW initial load on the AC sub-microgrid supplied by the thermal plant and the wind turbine. After 10 seconds, an additional 0.5 MW load is connected, totalling the connected load to 1.5 MW. The variation in AC load power is depicted in Fig. 16. At 20 seconds, the GFM inverter is connected to the system to enhance energy supply. By 30 seconds, the grid is connected, as illustrated in Fig. 17. Comparing the proposed GFM's frequency response to the conventional GFM when the grid is connected to the system reveals a grid connection overshoot for the conventional GFM, while the proposed GFM demonstrates no overshoot, as depicted in Fig. 17(a) and 17(b). Connection spikes pose risks to grid operations, potentially triggering grid protection systems and causing power disruptions. The proposed fault detection and synchronization controller mitigates generator and grid connection spikes, ensuring seamless microgrid operation. At 30 seconds, an additional 1 MW load is introduced into the system, bringing the total load to 3.5 MW. Despite the load increase, the power supplied by the GFM, thermal plant, and wind turbine remains stable, with excess power automatically directed to the grid. To validate the automatic power delivery of excess power to the grid, an additional 1 MW load is added at 50 seconds. Consequently, the GFM prioritizes power supply over other generators, as evidenced by the power sharing depicted in Fig. 18, while excess power is routed to the grid. The simulation continues until 60 seconds.

The power allocations of all generators and their synchronization, both for the conventional GFM and the proposed GFM, are illustrated in Fig. 17, while their corresponding frequency responses are depicted in Fig. 18. The load connection process described in Fig. 16 is mirrored in the frequency changes observed. At 40 seconds, all generators are interconnected, and an additional 1 MW load is introduced. Fig. 18 illustrates the synchronization of all generators and their stabilization of the synchronous frequency to the reference frequency of 60 Hz. This synchronization process demonstrates the coordinated operation of the generators, ensuring system stability and adherence to frequency standards.

Fig. 19 illustrates the DC link power change for both the conventional GFM and the proposed GFM on the DC side. In the conventional GFM control, power spikes of $10 \times 10^6\,W$ are observed during switching, as depicted in Fig. 19. This spike has the potential to trigger protection devices, posing operational risks. Conversely, the proposed GFM demonstrates seamless switching and activation, without such power spikes. Furthermore, it is noteworthy that the proposed GFM operates at a lower

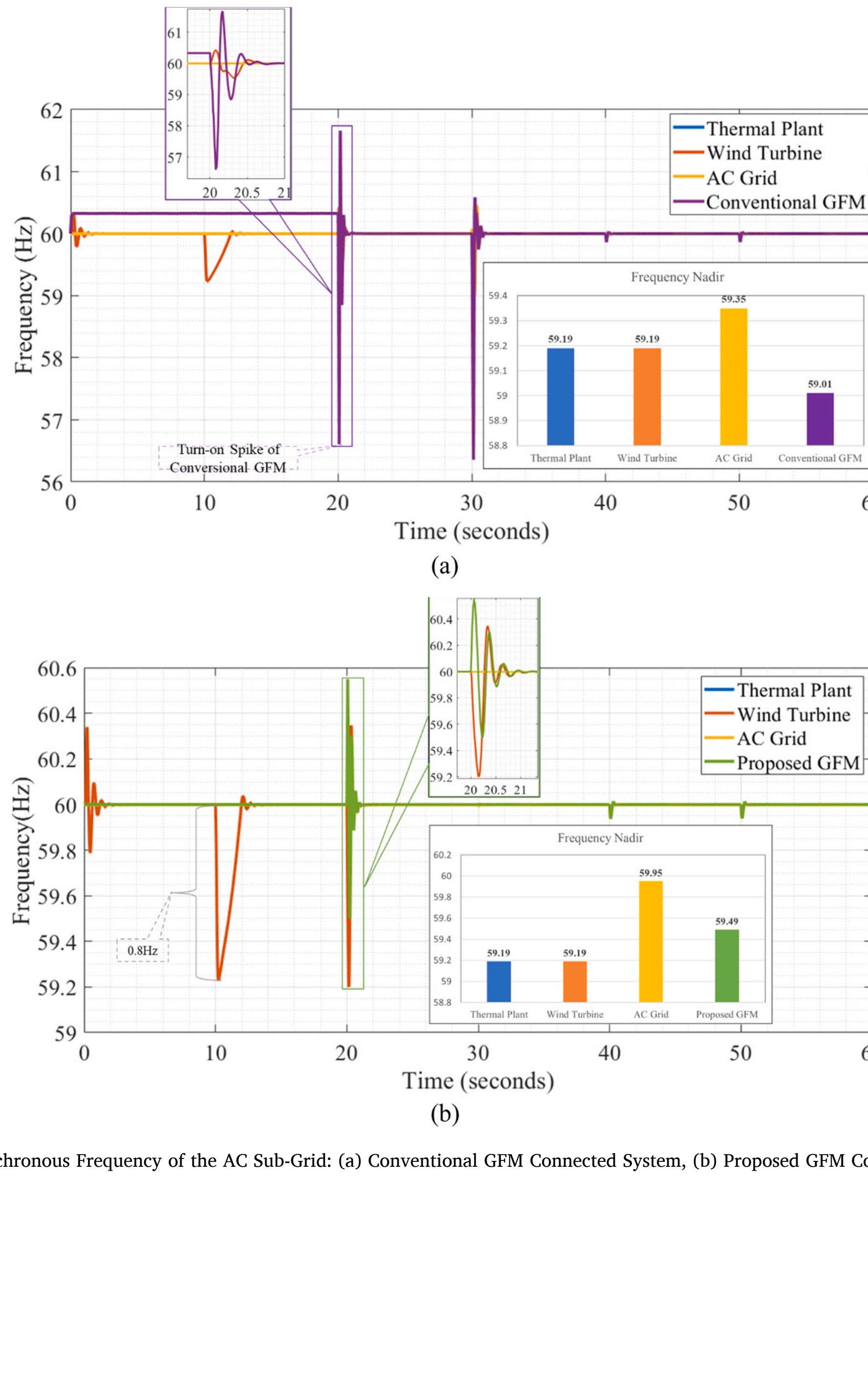


Fig. 18. Synchronous Frequency of the AC Sub-Grid: (a) Conventional GFM Connected System, (b) Proposed GFM Connected System.

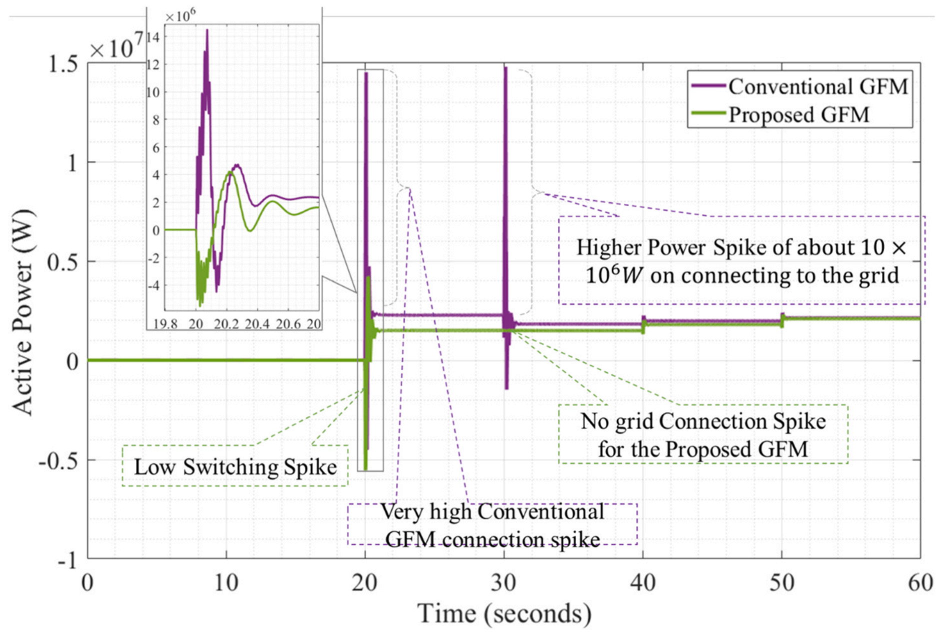


Fig. 19. Variation in DC Link Power During GFM Inverter Operation.

Table 3
Comparative Summary of the Conventional GFM and the Proposed GFM.

Situations	Parallel Operation	
	Conventional GFM	**Proposed GFM**
Frequency Nadir	59.75Hz	59.98Hz
Power Sharing for a step change	$6.9 \times 10^5 W$	$7.9 \times 10^5 W$
	Hybrid Microgrid Operation	
Power sharing Domination	No	Yes
Grid Connection Power Spike	$5 \times 10^6 W$	$2 \times 10^6 W$
Frequency Nadir	59.8Hz	59.95Hz
Grid Connection Freq. Spike	0.8 Hz	No Spike
	DC side Power response	
Grid Connection DC Power Spike	DC Power Spike of $10 \times 10^6 W$	DC No Power Spike
Connection Cases	**Synchronous Frequency Nadir**	
Case 1	High Spike	Low Spike
Case 2	59.75 Hz	59.98 Hz
Case 3	59.77 Hz	59.89 Hz
Case 4	59.01 Hz	59.49 Hz

DC power level of $1.6 \times 10^6 W$, attributed to the presence of a DC controller. This controller ensures precise power supply to the GFM, providing only the necessary power, compared to the conventional GFM operating at $1.8 \times 10^6 W$ without the DC controller. This difference underscores the efficiency and optimized operation facilitated by the proposed GFM's control mechanism on the DC side.

Table 3 provides a comparative summary of the conventional GFM and the proposed GFM, outlining key simulation operations and comparing the responses of both systems. Through this tabular summary, the effectiveness and control superiority of the proposed approach are evaluated. The results underscore the enhanced performance of the proposed GFM, indicating its control superiority. The findings suggest that effective DC link control for detection and synchronization is pivotal for achieving more stable operation. The superiority exhibited by the proposed GFM highlights the significance of robust control mechanisms in ensuring system stability and efficiency.

5. Conclusion

This paper addresses the challenges of detecting and synchronizing a hybrid DC/AC microgrid using a Grid-Forming (GFM) inverter. By examining both conventional and proposed GFM control strategies, we demonstrate the effectiveness of our approach in ensuring stability and seamless integration of diverse energy sources. Our simulations show that the proposed GFM controller significantly improves frequency nadir to 59.98Hz from 59.75Hz during a step increase, enhances power step response to $7.9 \times 10^5 W$ compared to $6.9 \times 10^5 W$ with the conventional controller, and eliminates the switching spike of $10 \times 10^5 W$ observed in the conventional GFM, thereby preventing potential DC overcurrent issues. These results underscore the proposed controller's superior performance in integrated fault detection and synchronization control, highlighting its potential to enhance the reliability of hybrid DC/AC microgrids. While the DC link fault detection control method for GFM offers substantial benefits, further research is needed to explore how DC bus fault detection can address the current limitations of GFM inverters. Overall, this research presents a promising control method for enhancing the stability and reliability of future sustainable power grids.

Funding

No funding have been received for this research

CRediT authorship contribution statement

Ryuto Shigenobu: Visualization, Validation, Supervision, Investigation, Formal analysis. **CHRISTIAN VERBE SUNJOH:** Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Masakazu Ito:** Visualization, Validation, Supervision, Methodology, Investigation, Conceptualization. **Hisao Taoka:** Visualization, Validation, Supervision, Methodology, Investigation. **Akiko Takahashi:** Visualization, Validation, Supervision, Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

Data will be made available on request.

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